
Why were 18 of 30 orders late?
Monday Orders (30 total)
1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20 21 22 23 24
25 26 27 28 29 30
Late (18)
On time (12)
Order Process (timings unknown)
Pick
Scan
Pack
Courier
Scanner Charging Checklist
Date
Scanner 1
Scanner 2
Mon
Tue
Wed

What problem are we solving?

Learn how to use AI to study a process problem: give the AI the process and the log, separate evidence from guesses, and plan a small test before spending money.

THE SITUATION

The problem

Sora manages a small online-shop warehouse. On Monday, 18 of 30 orders were shipped late, and customers started to complain. Her root problem is that late orders keep happening this week, and she does not know why. Her idea is to ask ChatGPT. She asked, "18 of 30 orders were late. Why? How do I fix it?" ChatGPT answered, "Your staff are too slow. Hire two more pickers and buy new scanners for $1,200." But Sora's own log shows that the two scanners were not charged at opening this week, and the charging checklist was blank. The AI guessed a cause and an expensive fix because the prompt gave it no evidence.

Today's goal and a three-step method

Learn how to use AI to study a process problem: give the AI the process and the log, separate evidence from guesses, and plan a small test before spending money.

STEP 1

Give the AI the process and the log

Paste the steps and the data. Tell the AI to use only them.

STEP 2

Separate evidence from guesses

Ask for late rates, the evidence, the unknowns, and possible causes in separate lists. Check the rates yourself.

STEP 3

Plan a small test

Before buying or hiring, test the most likely cause with one measure, one time period, and one limitation.

The order process

Fictional data created for this class. File: data/session_3_2_order_log.csv

CASE FACTS

The order process

- Every order goes through four steps: Pick, Scan, Pack, and hand to the Courier.
- Workers need a charged barcode scanner for the Scan step. The warehouse has two scanners.
- A charging checklist should be signed each evening. No timing data by step is available.

Order log

Fictional data created for this class. File: data/session_3_2_order_log.csv

Week	Day	Orders	Late orders	Scanners charged at opening	Checklist signed
Last week	Mon	30	2	2 of 2	Yes
Last week	Tue	28	3	2 of 2	Yes
Last week	Wed	31	2	2 of 2	Yes
Last week	Thu	29	1	2 of 2	Yes
Last week	Fri	32	2	2 of 2	Yes
This week	Mon	30	18	0 of 2	No
This week	Tue	29	12	0 of 2	No
This week	Wed	31	14	0 of 2	No

Five process words you need first

Before you ask an AI about a process problem, you must know these words. Each example uses Sora's warehouse.

KEY IDEA

Process step

One action in the work. The warehouse has four steps: Pick, Scan, Pack, and Courier.

KEY IDEA

Late rate

Late orders as a percent of all orders. This Monday: 18 / 30 = 60%.

KEY IDEA

Evidence

A fact in the log. Example: "This week, 0 of 2 scanners were charged at opening."

Five process words you need first (continued)

Before you ask an AI about a process problem, you must know these words. Each example uses Sora's warehouse.

KEY IDEA

Possible cause

An explanation that fits the evidence but is not proven. Example: "Uncharged scanners slowed the Scan step."

KEY IDEA

Small test

A short, low-cost check of one possible cause. Example: sign the charging checklist every day for one week and compare the late rate.

Why the AI answer was wrong

"Sounds like a solution" and "fits the evidence" are different questions. Compare the AI answer with Sora's log.

What the AI said	What the log shows	Problem
Staff are too slow	No timing data by step	A guess about people
Hire two more pickers	Orders per day are the same as last week (about 30)	Not supported
Buy new scanners for $1,200	The scanners work; they were not charged	An expensive fix for a guessed cause

KEY POINTS

What the table shows

- Last week, 10 of 150 orders were late: 6.7%. This week, 44 of 90 orders were late: 48.9%.
- The big change this week is that the scanners were not charged and the checklist was not signed.
- This pattern points to a possible cause. It does not prove it. Something else may also have changed this week.

How the three-step method fixes the answer

The instructor runs Poor Prompt 1 and Improved Prompts 1 and 2 below in front of the class and compares the answers.

KEY IDEA

Step 1 fixes the missing-evidence problem

With the process and the log, the AI can compare this week with last week.

KEY IDEA

Step 2 fixes the blame problem

Separate lists keep facts, unknowns, and guesses apart, so no worker is blamed without evidence.

KEY IDEA

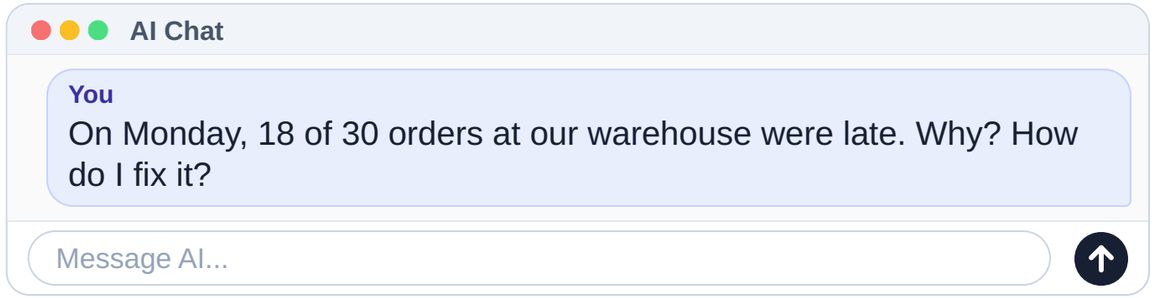
Step 3 fixes the expensive-fix problem

A one-week checklist test costs almost nothing. Sora should run it before spending $1,200.

Poor prompt 1

Same case. Compare what the prompt asks for.

POOR PROMPT 1 - DO NOT USE



WHAT THE AI OFTEN ANSWERS

A typical result

"Late orders usually mean your staff are too slow or understaffed. Hire two more pickers and buy new, faster scanners."

WHY IT IS WEAK

Reasons

- It gives no process and no log, so the AI uses general reasons for late orders.
- It asks "why" and "how to fix" in one step, so the AI jumps to a solution without evidence.
- The result blames workers and suggests spending money on a guessed cause.

Poor prompt 2

Same case. Compare what the prompt asks for.

POOR PROMPT 2 - DO NOT USE

AI Chat

You

Our warehouse is slow. This week our two scanners were not charged.
Tell me which worker is responsible and how to punish them.

Message AI...

WHAT THE AI OFTEN ANSWERS

A typical result

A list of "accountability steps," such as warnings for the closing staff, without any check of the data.

WHY IT IS WEAK

Reasons

- It asks the AI to blame a person, but the data does not say who forgot to charge the scanners.
- It treats one possible cause as a proven fact.
- Blame does not fix the process. A missing checklist step is a process problem.

Improved prompt 1

Steps 1 and 2. Copy this prompt exactly.

IMPROVED PROMPT 1 - COPY AND SEND

AI Chat

You

I manage a small warehouse. Use only the process and the log below.

Process: every order goes through Pick, Scan, Pack, and Courier.
Workers need a charged barcode scanner for the Scan step. We have two scanners.
A charging checklist should be signed each evening. No timing data by step is available.

Log (orders, late orders, scanners charged at opening, checklist signed):
Last week: Mon 30, 2, 2 of 2, Yes; Tue 28, 3, 2 of 2, Yes; Wed 31, 2, 2 of 2, Yes;
Thu 29, 1, 2 of 2, Yes; Fri 32, 2, 2 of 2, Yes.
This week: Mon 30, 18, 0 of 2, No; Tue 29, 12, 0 of 2, No; Wed 31, 14, 0 of 2, No.

1. Calculate the late rate (late orders / orders) for each week.
2. List: Evidence, Unknowns, and Possible causes, in three separate lists.
Show the calculation for this week's late rate so I can check it.
Do not blame individual workers. Do not recommend purchases or hiring.

Message AI...

Why it works: improved prompt 1

Each part of the prompt matters.

WHY IT WORKS

Reasons

- It gives the process and the full log, which fixes the missing evidence in Poor Prompt 1.
- Asking for late rates turns "many orders were late" into numbers you can compare.
- Three separate lists keep facts, unknowns, and guesses apart.
- "Do not blame individual workers" prevents the problem in Poor Prompt 2.
- "Do not recommend purchases or hiring" stops the expensive fix before the cause is tested.

Improved prompt 2

Step 3, sent in the same chat after Improved Prompt 1. Copy this prompt exactly.

IMPROVED PROMPT 2 - COPY AND SEND

AI Chat

You

I checked your rates: last week 6.7% (10 of 150), this week 48.9% (44 of 90).
The most likely possible cause is uncharged scanners.

Plan a one-week test of this cause. Include:
- The one change we will make
- The one measure we will record each day
- What result would support the cause, compared with 6.7% and 48.9%
- One reason the result can suggest the cause but not prove it

Keep the plan under 120 words. Do not include any purchase.

Message AI...

Why it works: improved prompt 2

Each part of the prompt matters.

WHY IT WORKS

Reasons

- It gives the checked rates, so the plan compares with real numbers.
- One change and one measure keep the test simple and clear.
- Asking what result would support the cause sets the target before the test.
- The last item makes the AI state the limit of the test honestly.

Exercise 1: Find the evidence

Guided practice. Follow the steps in order.

GUIDED PRACTICE

Steps

1. Open ChatGPT and start a **new chat**. Copy Poor Prompt 1 from the box above, paste it, and send it. Write down the cause and the fix the AI suggested. You will need them for your report.
2. Start another **new chat**, so the first answer does not affect the second. Copy Improved Prompt 1, paste it, and send it.
3. Check the late rates with a calculator: last week 10 / 150 = 6.7%, and this week 44 / 90 = 48.9%. Compare the three AI lists with the answer key below.
4. In the same chat, copy Improved Prompt 2, paste it, and send it. Check that the plan has one change, one daily measure, and one limitation, and no purchase.
5. For the Results section of your report, write the AI answer from Step 1 and one sentence that explains which part of it was not supported by the log.

Answer key

Use this table to check the AI.

List	Items
Late rate	Last week 6.7% (10 of 150). This week 48.9% (44 of 90).
Evidence	This week, 0 of 2 scanners were charged at opening on all three days, and the checklist was not signed. Last week, both scanners were charged every day and the late rate was low. Orders per day are about the same in both weeks.
Unknowns	Time spent in each step; whether anything else changed this week (staff, courier pickup time, order size)
Possible causes	Uncharged scanners slowed the Scan step; another change this week that is not in the log

Exercise 2: Read the test results

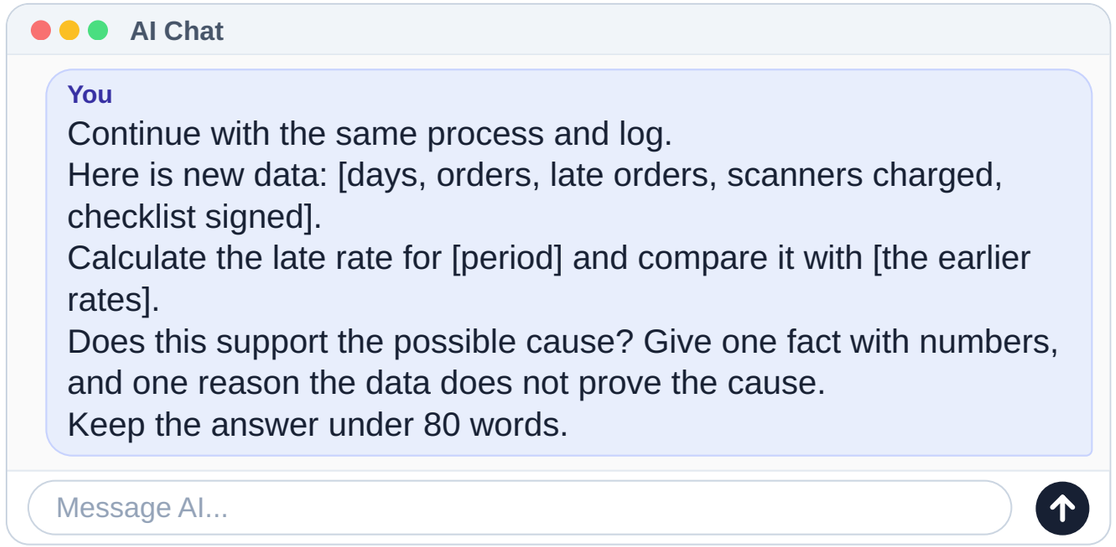
Applied practice with a template.

HOW IT WORKS

Read first

- Sora restarted the charging checklist on Thursday. Now she needs to read new data carefully. Use this template. Replace each part in square brackets.

TEMPLATE - FILL IN, THEN SEND



Part A: Worked example (run it)

Copy and send this completed prompt in the same chat.

PART A - COPY AND SEND

AI Chat

You

Continue with the same process and log.
Here is new data: This week Thu 30 orders, 3 late; Fri 32 orders, 2 late.
On both days, 2 of 2 scanners were charged and the checklist was signed.
Calculate the late rate for Thursday and Friday together and compare it
with last week (6.7%) and with Monday to Wednesday (48.9%).
Does this support the possible cause? Give one fact with numbers, and one reason the data does not prove the cause.
Keep the answer under 80 words.

Message AI...

CHECK THE ANSWER

Expected result

Thursday and Friday had 5 late orders out of 62, so the late rate is 5 / 62 = 8.1%. This is close to last week's 6.7% and far below 48.9%, so it supports the possible cause. It does not prove it, because two days is a short period and other things may have changed, too.

Part B: Your turn (write it)

Fill in the template yourself and send it.

YOUR TASK

Write your own prompt

The next week, the checklist was signed every day and both scanners were charged. Results: Mon 30 orders, 2 late; Tue 29, 3 late; Wed 31, 2 late; Thu 30, 1 late; Fri 28, 2 late. Sora also notes that a new worker started on Tuesday. Fill in the template yourself and send it. Your prompt must include all five days and the note about the new worker.

CHECK THE ANSWER

Expected result

The late rate is 10 / 148 = 6.8%, about the same as last week's 6.7%. This strongly supports the possible cause. A good answer still names a limitation: the new worker also started that week, so the result cannot prove that charging alone fixed the problem.

Write your session report

Use Report_template.docx in Microsoft Word. Save it as FirstName_StudentID_W10_Keyword.docx and upload it to iLearn by 23:59 on class day.

SUBMIT

One report for this session

- **Introduction:** Explain Sora's root problem (many late orders this week), her idea to ask AI, and why the first AI answer was not useful.
- **Method:** Copy Poor Prompt 1, Improved Prompts 1 and 2, the Part A prompt, and the Part B prompt you wrote.
- **Results:** Include the AI answer to Poor Prompt 1, the late rates with your calculator check, the three lists from Improved Prompt 1, the test plan, and your late-rate checks for Part A and Part B.
- **Conclusion:** Recommend what Sora should do next, in one or two sentences. Name one fact that supports it and one limitation of the evidence.

How to know the work is complete

Check your report before you hand it in.

VERIFY

Success check

- You can explain in one sentence why "hire two pickers and buy new scanners" was not supported by the log, and your Part B prompt produces a late rate of 6.8% that you checked with a calculator.